

Muskan Pardasani

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EDUCATION

University of Wisconsin, Madison

Aug 2023 - May 2025

Degree: *Masters of Science in Data Science (3.73/4.0)*

Relevant courses: Probability Theory, Statistical Models & Methods, Optimization, Database Management, Adv. NLP, HCI

Indian Institute of Technology, Indore, India

July 2018 - June 2022

Degree: *Bachelor of Technology (Electrical Engineering) (8.2/10)*

Relevant courses: Data Structures & Algorithms, Object Oriented Programming (C++), Calculus, Probability & Statistics

WORK EXPERIENCE

C3 AI- Data Scientist Intern (Generative AI, LLMs)

June 2024 – Aug 2024

Redwood City, California

- Designed and productionized an LLM-based pipeline using Python and OpenAI API, boosting client-model accuracy from **16% to 92%**.
- Developed a production-ready generative AI model for sugar industry clients, **increasing productivity by 50%**, enabling multi-industry expansion through **transformer-based architectures and retrieval-augmented generation (RAG)**.
- Leveraged **CUDA and PyTorch** to accelerate LLM inference pipelines, optimizing token-level latency and throughput across large **10B+ parameter models**; benchmarked performance under real-time constraints.

Bosch- AI/ML Software Engineer

Aug 2022 – June 2023

Bengaluru, India

- Designed and implemented a real-time recommendation microservice (**Python, scikit-learn, Flask, Docker, Kubernetes**) serving **1M+ daily requests** and increasing user engagement by around **25%**.
- Integrated **ML pipelines** using Docker and AWS, and accelerated experimentation using **GPU-backed EC2 instances**.
- Collaborated with cross-functional R&D teams to integrate predictive-analytics dashboards into Bosch's central platform, delivering actionable insights to **125+ global teams**.

Graduate Teaching Assistant, Physics

Jan 2025- May 2025

University of Wisconsin, Madison

- Teaching Assistant for 2 different physics courses.** Lead weekly discussions and labs for **50+ students**, simplifying complex **electrostatics & magnetics** concepts and **physics in real life** through interactive problem-solving.
- Design quizzes and worksheets that reinforce fundamental and advanced techniques.

ACHIEVEMENTS

- Google CodeJam 2022:** Achieved a **Global Rank 61 (top 0.6%)** and an **All India Rank 19** amongst **10,000+ candidates** in Google's CodeJam for Women, a global competitive programming event.
- Amazon Machine Learning Challenge 2021:** Ranked in the **top 6% (All India Rank 248)** among 4,000+ teams.
- Co-Founder & the First President, The Consulting, Finance, & Analytics Club, IIT Indore (2020-21)**
- IIT Entrance Examination (JEE):** Ranked in the **top 0.02% - All India Rank 484** amongst **2.4 Million candidates** (Mains), and an **All India Rank 5244** amongst **250,000 candidates** (Advanced) in 2018.

TECH SKILLS

Programming: C, C++, Python, Java, SQL, MATLAB, R, Javascript

Skills/Frameworks: Pytorch, Tensorflow, Tableau, Predictive Modeling, CUDA, AWS, Docker, OpenAI Gym

RESEARCH PROJECTS

Off Policy Evaluation for Fitted Q Evaluation (Reinforcement Learning) (with Prof. Josiah Hanna)

Feb 2025 – Ongoing

- Implemented **Fitted Q-Evaluation (FQE)** to assess policy performance in **off-policy evaluation (OPE)**.
- Exploring option-based FQE, extending traditional FQE to hierarchical decision-making structures. Analyzing preliminary results to validate the effectiveness of this approach in policy assessment.

Minirel: A Mini Relational Database System (Course Project – CS 564 Database Management)

Sept 2024 – Dec 2024

- Designed and implemented a mini relational database system (Minirel) from scratch**, following database architecture principles, including buffer management, indexing, and query processing.
- Built a **query execution engine** supporting **SQL-like operations**, including selection, projection, and join algorithms.
- Designed a storage manager for **disk-based data organization** and optimized page replacement with LRU and Clock algorithms. (**C++, File I/O, Data Structures**)

Multimodal Model Adaptation for Medical Imaging (Course Project- CS 769 Adv NLP)

Sept 2024 – Dec 2024

- Achieved a **63% accuracy boost** (from CLIP **49.89%** to CoOp **81.47%**) by leveraging **lightweight adaptation techniques** such as **prompt tuning (CoOp)** and **CLIP Adapter**, proving their effectiveness over domain-specific pretraining.
- Demonstrated that **prompt tuning (CoOp) outperforms multimodal fine-tuning**, revealing that **textual alignment is key** for domain adaptation. CLIP Adapter improved accuracy (**77.80%-78.03%**), but lacked in vision-language adaptation.